

Utility of *PAX1*/*JAM3* methylation analysis for triage of high-risk HPV-positive individuals

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Abstract

Introduction: We sought to assess the clinical utility of methylation detection of paired box gene 1 (*PAX1*) and junctional adhesion molecule 3 (*JAM3*) in the triage of individuals testing positive for high-risk human papillomavirus (HPV).

Methods: Cervical secretions from 312 high-risk HPV-positive patients were analyzed for dual-gene methylation of *PAX1* and *JAM3* (*PAX1m/JAM3m*). Methylation levels were compared across histologically confirmed cervical lesions. Using histopathology as the reference standard, the triage performance of *PAX1m/JAM3m* was evaluated against cytology and HPV-16/18 genotyping.

Results: Methylation positivity increased in statistical significance with lesion severity ($P < .001$ for trend). For the detection of cervical intraepithelial neoplasia (CIN) 2 or more severe lesions (CIN2+), *PAX1m/JAM3m* yielded a sensitivity of 91.8% (95% CI, 84.1%-96.2%), specificity of 90.7% (95% CI, 85.7%-94.1%), and an area under the receiver operating characteristic curve of 0.912 (95% CI, 0.874-0.951), outperforming cytology, HPV-16/18 genotyping, and their combinations. Using *PAX1m/JAM3m* positivity as a criterion for colposcopy referral, 1 CIN2+ case was detected per 1.22 referrals, reducing the colposcopy referral rate by approximately 19.2% and increasing the CIN2+ detection rate by 39.4% compared with cytology at the atypical squamous cells of undetermined significance threshold.

Discussion: *PAX1m/JAM3m* levels are strongly associated with cervical lesion severity and represent a promising triage strategy for high-risk HPV-positive individuals.

Key words: cervical cancer, DNA methylation, *PAX1*, *JAM3*, triage performance, colposcopy referrals

Introduction

Human papillomavirus (HPV) is a mucosotropic virus comprising more than 200 identified subtypes and can induce benign or malignant proliferative lesions in epithelial and mucosal tissues.¹ Persistent infection with high-risk HPV is a major etiologic factor in the development of cervical cancer. Given its high sensitivity and strong negative predictive value (NPV), high-risk HPV nucleic acid testing is currently recommended as the primary screening modality for cervical cancer.² A positive high-risk HPV result, however, does not necessarily indicate the presence of cervical cancer. Approximately 80% of women acquire an HPV infection during their lifetime, the majority of which resolve spontaneously, and only a small subset of persistent high-risk HPV infections progresses to cervical cancer.³ The high false-positive rate associated with high-risk HPV testing may result in psychological burden, overdiagnosis, and overtreatment among screened individuals.⁴ Cytology is commonly employed as a triage method for high-risk HPV-positive cases; however, its sensitivity and specificity are suboptimal and can be affected by sample quality, slide preparation, patient age, and the technical expertise of laboratory personnel.⁵ Therefore, optimizing triage

strategies for high-risk HPV-positive individuals remains a critical clinical need.

DNA methylation, a key epigenetic modification, regulates gene expression by promoting gene silencing or activation and plays an essential role in tumor initiation, progression, and metastasis.^{6,7} Paired box gene 1 (*PAX1*) and junctional adhesion molecule 3 (*JAM3*) are tumor-associated genes whose promoter methylation has been closely linked to the development of cervical precancerous lesions.^{8,9} *PAX1*, a member of the paired box gene family, functions as a tumor suppressor by regulating transcription.¹⁰ Promoter hypermethylation of *PAX1* leads to its transcriptional silencing, thereby diminishing its tumor-suppressive activity and facilitating carcinogenesis.¹¹ Although the methylation frequency of *PAX1* is negligible in normal cervical tissue (0%), it increases markedly in cervical cancer, reaching 94.4% ($P < .001$).¹² *JAM3*, a member of the junctional adhesion molecule family, is involved in cell adhesion and signal transduction. It has been shown to suppress cancer cell proliferation both in vitro and in vivo by inhibiting Wnt signaling. Methylation of the *JAM3* promoter leads to its downregulation, contributing to tumor

progression.¹³ In cervical cancer, *JAM3* methylation (*JAM3m*) has emerged as a promising biomarker, correlating with precancerous lesions, advanced International Federation of Gynecology and Obstetrics stages, and lymph node metastasis.¹⁴ Given these findings, methylation detection of *PAX1* and *JAM3* may serve as a valuable biomarker-based approach for early cervical cancer screening and risk stratification. This study aimed to evaluate the utility of dual-gene methylation testing (*PAX1m/JAM3m*) in high-risk HPV-positive individuals, with the goal of informing triage strategies for patients with abnormal cervical cancer screening results.

Methods

General information

This study was conducted with individuals who had positive high-risk HPV test results from cervical cancer screenings performed between January 2024 and June 2025. Inclusion criteria were (1) aged 18 years or older, (2) detection of 1 or more high-risk HPV subtypes in cervical secretions, and (3) completion of both liquid-based cytology and cervical histopathologic examination. Exclusion criteria were (1) a history of genital tract malignancies or other tumors; (2) treatment for cervical or vaginal lesions within the preceding 6 months; (3) autoimmune disorders or on current immunosuppressive therapy; and (4) conditions known to influence DNA methylation, such as pregnancy, lactation, or HIV infection. The study protocol was approved by the Medical Ethics Committee of Ziyang Central Hospital (approval No. 2023331).

Cervical secretion collection and high-risk HPV DNA testing

Cervical secretions were collected using a disposable cytobrush, placed into sample collection tubes, and promptly submitted for analysis. High-risk HPV DNA testing was performed using the High-Risk Human Papillomavirus Genotyping Nucleic Acid Detection Kit (Shanghai Zhijiang Biotechnology Co, Ltd). Remaining samples were stored at -70°C . Detection of any of HPV-16, HPV-18, HPV-31, HPV-33, HPV-35, HPV-39, HPV-45, HPV-51, HPV-52, HPV-56, HPV-58, HPV-59, HPV-66, or HPV-68 was classified as high-risk HPV positive.

PAX1 and *JAM3* gene methylation detection

Residual DNA samples from high-risk HPV testing were subjected to *PAX1* and *JAM3* methylation analysis using the Human *PAX1* and *JAM3* Gene Methylation Detection Kit (polymerase chain reaction fluorescence probe method; Beijing Origin Polybio Co, Ltd) in accordance with the manufacturer's instructions. The ΔCt value was calculated as follows: $\Delta\text{Ct} = \text{Ct}_{\text{target gene}} - \text{Ct}_{\text{reference gene (GAPDH)}}$. Methylation was considered positive if $\Delta\text{Ct}_{\text{PAX1}} \leq 6.6$ or $\Delta\text{Ct}_{\text{JAM3}} \leq 10.0$, thresholds recommended by the kit manufacturer.

Data collection

Demographic and clinical data, including age, medical history, liquid-based cytology, and cervical histopathology results, were extracted from the hospital's electronic health record and laboratory information systems. Liquid-based cytology

results were reported by pathologists in accordance with the 2014 revised Bethesda System for cervical cytology.

Data analysis

Statistical analyses were conducted using SPSS, version 25.0, software (IBM) and GraphPad Prism, version 9, software (GraphPad Software). Continuous variables with normal distribution were expressed as mean (SD) and compared using *t* tests or analysis of variance. Non-normally distributed continuous variables were presented as median (IQR) and analyzed using the Kruskal-Wallis test. Categorical variables were expressed as counts (percentages) and compared using the χ^2 test. Trends in methylation levels across lesion grades were evaluated using the Jonckheere-Terpstra test. Using histopathologic diagnosis as the reference standard, diagnostic performance metrics, including sensitivity, specificity, positive predictive value (PPV), NPV, and 95% CIs, were calculated for each detection method in identifying cervical intraepithelial lesion 2 or more severe (CIN2+) lesions. $P < .05$ was considered statistically significant.

Results

Baseline characteristics

A total of 312 high-risk HPV-positive individuals (mean [SD] age, 49 [14] years) were enrolled: 194 (62.2%) with cervicitis, 20 (6.4%) with (CIN) 1, 20 (6.4%) with CIN2, 31 (9.9%) with CIN3, 37 (11.9%) with squamous cell carcinoma (SCC), and 10 (3.2%) with adenocarcinoma. Among these, 202 individuals (64.7%) were negative and 110 (35.3%) were positive for *PAX1m/JAM3m*. Statistically significant differences were observed among groups with respect to age, HPV-16/18 genotyping, liquid-based cytology results, and histopathologic diagnoses (Table 1).

Correlation analysis of methylation levels across cervical lesions

The positivity rates for *PAX1m* in cervicitis, CIN1, CIN2, CIN3, SCC, and adenocarcinoma were 5.2% (10/194), 25.0% (5/20), 70.0% (14/20), 83.9% (26/31), 100.0% (37/37), and 100.0% (10/10), respectively; for *JAM3m*, the corresponding rates were 6.2% (12/194), 10.0% (2/20), 50.0% (10/20), 87.1% (27/31), 100.0% (37/37), and 100.0% (10/10), respectively; and for *PAX1m/JAM3m*, the rates were 7.7% (15/194), 25.0% (5/20), 75.0% (15/20), 90.3% (28/31), 100.0% (37/37), and 100.0% (10/10), respectively (Figure 1A). Both *PAX1m* and *JAM3m* detected all cases of cervical cancer.

Compared with the cervicitis group, *PAX1m* and *JAM3m* levels were clinically significantly elevated in the CIN2, CIN3, SCC, and adenocarcinoma groups ($P < .001$). Jonckheere-Terpstra trend analysis confirmed a statistically significant increase in methylation levels with increasing lesion severity ($P < .001$ for trend). For *PAX1m*, no statistically significant difference was observed between normal tissue and CIN1 ($P = .13$), whereas statistically significant differences were noted between CIN1 and CIN2 ($P = .01$), CIN2 and CIN3 ($P = .02$), and CIN3 and SCC ($P = .001$) (Figure 1B). For *JAM3m*, no statistically significant difference was observed between normal tissue and CIN1 ($P = .24$); however, statistically significant differences were observed between CIN1 and

Table 1 Baseline characteristics of the study population.

Characteristic	Total (n = 312)	PAX1m/JAM3m negative (n = 202)	PAX1m/JAM3m positive (n = 110)	Test statistic	P value
Age, mean (SD), y	49 (14)	46 (12)	54 (14)	5.503	<.001
High-risk HPV genotype, No. (%)				21.804	<.001
HPV-16/18 positive	111 (35.6)	53 (26.2)	58 (52.7)		
HPV other 12 positive	201 (64.4)	149 (73.8)	52 (47.3)		
Liquid-based cytology, No. (%)				18.476	<.001
No intraepithelial lesion or malignancy	142 (45.5)	110 (54.5)	32 (29.1)		
≥AS-CUS	170 (54.5)	92 (45.5)	78 (70.9)		
Pathology, No. (%)				41.328	<.001
Cervicitis	194 (62.2)	179 (88.6)	15 (13.6)		
CIN1	20 (6.4)	15 (7.4)	5 (4.6)		
CIN2	20 (6.4)	5 (2.5)	15 (13.6)		
CIN3	31 (9.9)	3 (1.5)	28 (25.5)		
SCC	37 (11.9)	0	37 (33.6)		
Adenocarcinoma	10 (3.2)	0	10 (9.1)		

Abbreviations: AS-CUS, atypical squamous cells of undetermined significance; CIN, cervical intraepithelial neoplasia; HPV-16/18, human papillomavirus 16 or 18; HPV other 12, any of the 12 additional high-risk human papillomavirus genotypes; PAX1m/JAM3m, dual-gene methylation of PAX1 or JAM3; SCC, squamous cell carcinoma.

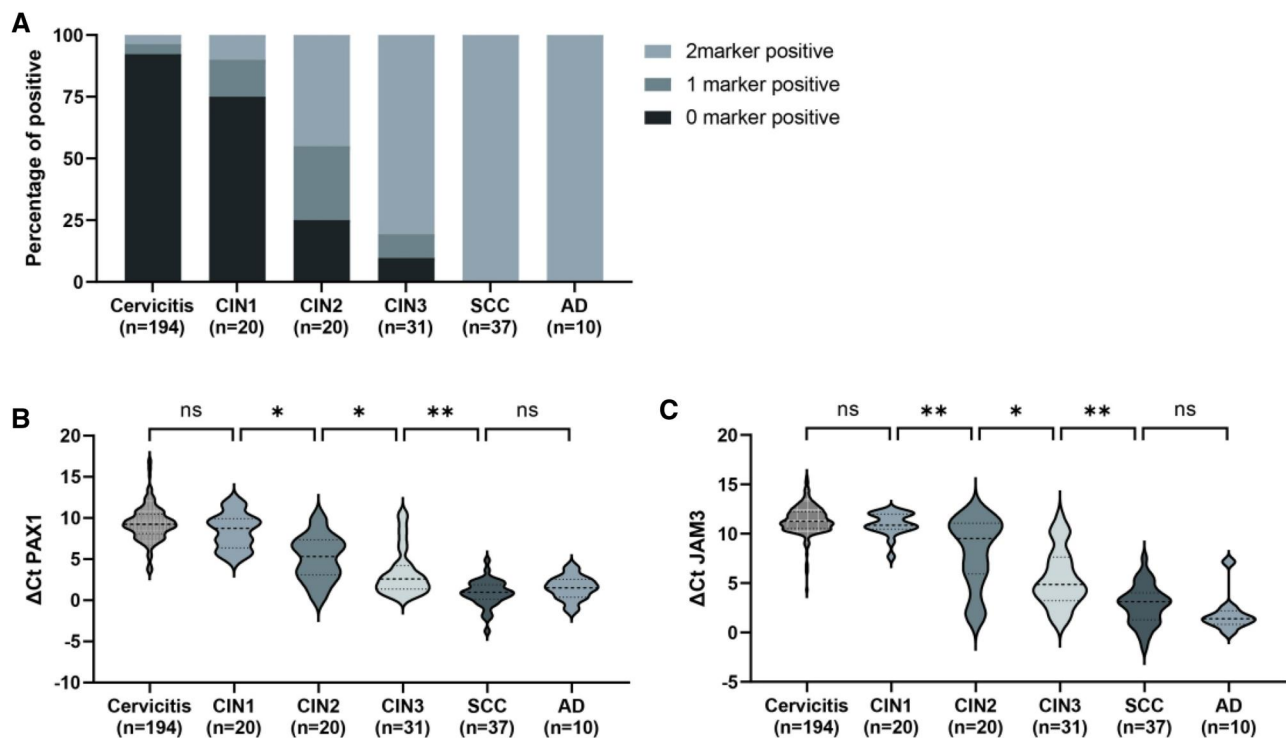


Figure 1 (A) PAX1m/JAM3m positivity rates across histologically confirmed cervical lesion grades. (B) Distribution of PAX1 methylation levels across cervical lesion grades. (C) Distribution of JAM3 methylation levels across cervical lesion grades. $\Delta Ct = Ct_{\text{target gene}} - Ct_{\text{reference gene}}$. AD indicates adenocarcinoma; CIN, cervical intraepithelial neoplasia; ns, not significant; PAX1m/JAM3m, dual-gene methylation of PAX1 and JAM3; SCC, squamous cell carcinoma. * $P < .05$. ** $P < .01$.

CIN2 ($P = .005$), CIN2 and CIN3 ($P = .02$), and CIN3 and SCC ($P = .001$) (Figure 1C).

Clinical performance of methylation, HPV-16/18 genotyping, and cytology in diagnosing CIN2+ lesions

Using histopathologic diagnosis as the reference standard, PAX1m/JAM3m achieved a sensitivity of 91.8% (95% CI, 84.1%-96.2%), a specificity of 90.7% (95% CI, 85.7%-94.1%), a PPV of 81.8% (95% CI, 73.1%-88.3%), an NPV of 96.0% (95% CI, 92.1%-98.1%), and an area under the curve (AUC)

of 0.912 (95% CI, 0.874-0.951), outperforming both cytology (liquid-based cytology $\geq$ atypical squamous cells of undetermined significance [AS-CUS]) and HPV-16/18 genotyping (Table 2).

Combining PAX1m/JAM3m with cytology or HPV-16/18 genotyping, positivity for any of PAX1m/JAM3m, HPV-16/18, or liquid-based cytology $\geq$ AS-CUS yielded the highest sensitivity for CIN2+ lesions (98.0% [95% CI, 92.1%-99.6%]) and NPV (97.2% [95% CI, 89.3%-99.5%]), albeit with reduced specificity and PPV. In contrast, simultaneous positivity for

Table 2 Clinical performance comparison of *PAX1*/*JAM3* methylation and other methods for diagnosing CIN2+ lesions.

Method	Sensitivity, % (95%CI)	Specificity, % (95%CI)	PPV, % (95%CI)	NPV, % (95%CI)	AUC (95% CI)
<i>PAX1m</i> / <i>JAM3m</i> positive	91.8 (84.1-96.2)	90.7 (85.7-94.1)	81.8 (73.1-88.3)	96.0 (92.1-98.1)	0.912 (0.874-0.951)
<i>PAX1m</i> positive	88.8 (80.4-94.0)	93.0 (88.5-95.9)	85.3 (76.6-91.3)	94.8 (90.6-97.2)	0.909 (0.868-0.950)
<i>JAM3m</i> positive	85.7 (76.9-91.7)	93.5 (89.0-96.2)	85.7 (76.9-91.7)	93.5 (89.0-96.2)	0.896 (0.851-0.941)
Liquid-based cytology ≥ AS-CUS	73.5 (63.4-81.6)	54.2 (47.3-61.0)	42.4 (34.9-50.2)	81.7 (74.1-87.5)	0.638 (0.573-0.703)
HPV-16/18 positive	55.1 (44.7-65.1)	73.4 (66.8-79.1)	48.6 (39.1-58.3)	78.1 (71.6-83.5)	0.642 (0.575-0.710)
<i>PAX1m</i> / <i>JAM3m</i> positive or liquid-based cytology ≥ AS-CUS	95.9 (89.3-98.7)	50.9 (44.0-57.8)	47.2 (40.2-54.4)	96.5 (90.6-98.9)	0.734 (0.680-0.789)
<i>PAX1m</i> / <i>JAM3m</i> positive and liquid-based cytology ≥ AS-CUS	67.3 (57.0-76.3)	95.3 (91.3-97.6)	86.8 (76.7-93.2)	86.4 (81.2-90.4)	0.813 (0.754-0.873)
<i>PAX1m</i> / <i>JAM3m</i> positive or HPV-16/18 positive	94.9 (87.9-98.1)	67.3 (60.5-73.4)	57.1 (49.1-64.7)	96.6 (91.9-98.8)	0.811 (0.763-0.859)
<i>PAX1m</i> / <i>JAM3m</i> positive and HPV-16/18 positive	52.0 (41.8-62.2)	96.7 (93.1-98.6)	87.9 (76.1-94.6)	81.5 (76.1-86.0)	0.744 (0.677-0.810)
<i>PAX1m</i> / <i>JAM3m</i> positive or liquid-based cytology ≥ AS-CUS or HPV-16/18 positive	98.0 (92.1-99.6)	32.2 (26.1-39.0)	39.8 (33.7-46.3)	97.2 (89.3-99.5)	0.651 (0.591-0.711)
<i>PAX1m</i> / <i>JAM3m</i> positive and liquid-based cytology ≥ AS-CUS and HPV-16/18 positive	34.7 (25.5-45.1)	99.5 (97.0-100.0)	97.1 (83.4-99.9)	76.9 (71.4-81.6)	0.671 (0.600-0.742)

Abbreviations: AS-CUS, atypical squamous cells of undetermined significance; AUC, area under the curve; CIN, cervical intraepithelial lesion; HPV, human papillomavirus; *JAM3m*, *JAM3* methylation; *JAM3m* positive, $\Delta Ct_{JAM3} \leq 10.0$; NPV, negative predictive value; *PAX1m*, *PAX1* methylation; *PAX1m* positive, $\Delta Ct_{PAX1} \leq 6.6$; *PAX1m*/*JAM3m*, dual-gene methylation for *PAX1* or *JAM3*; PPV, positive predictive value.

PAX1m/*JAM3m*, HPV-16/18, and liquid-based cytology ≥ AS-CUS yielded the highest specificity (99.5% [95% CI, 97.0%-100.0%]) and PPV (97.1% [95% CI, 83.4%-99.9%]). Receiver operating characteristic analysis indicated that combining *PAX1m*/*JAM3m* with cytology or HPV-16/18 genotyping did not further improve diagnostic performance for CIN2+ lesions (Table 2, Figure 2).

Utility of *PAX1m*/*JAM3m* in guiding colposcopy referral
Among all high-risk HPV-positive individuals ($n = 312$), using *PAX1m*/*JAM3m* positivity as the criterion for colposcopy referral identified 1 CIN2+ case per 1.22 referrals. This strategy increased the immediate detection rate of CIN2+ lesions by 33.2% compared with HPV-16/18 genotyping and reduced the colposcopy referral rate by approximately 19.2% relative to liquid-based cytology ≥ AS-CUS while increasing the CIN2+ detection rate by 39.4% (Table 3). In the HPV 12 other-positive subgroup, *PAX1m*/*JAM3m* reduced the colposcopy referral rate by approximately 34.3% and increased the CIN2+ detection rate by 44.4% compared with liquid-based cytology ≥ AS-CUS (Table 3).

Discussion

Cervical cancer remains 1 of the most prevalent gynecologic malignancies, posing a substantial threat to women’s health, particularly in geographic regions with limited medical resources. According to the National Cancer Center’s 2024 report, China documented 150 700 new cases of cervical cancer in 2022, ranking it as the fifth-most common malignancy among women in China.¹⁵ Persistent infection with high-risk HPV is a key etiologic factor in the progression from CIN to invasive carcinoma. Owing to its high sensitivity, high-risk HPV

nucleic acid testing has become a standard screening tool, but high-risk HPV DNA testing detects only the presence of infection, offering no information about infection persistence or lesion severity. Current guidelines recommend HPV-16/18 genotyping and liquid-based cytology for the triage of high-risk HPV-positive individuals, with positive findings prompting colposcopy referral.² Nonetheless, studies have shown that among high-risk HPV-positive individuals with cytologic findings of AS-CUS or worse, the immediate prevalence of high-grade lesions is only 12.2%.¹⁶ This finding leads to excessive colposcopy referrals, contributing to patient anxiety and straining health care resources.¹⁷ In addition, the subjectivity inherent in cytologic interpretation compromises reproducibility.¹⁸ These limitations underscore the urgent need for objective, high-performance biomarkers to improve risk stratification in high-risk HPV-positive populations.

DNA methylation, a key epigenetic mechanism, is frequently dysregulated in cancer and remains stable throughout neoplastic progression. The detection of gene-specific methylation patterns holds promise for early diagnosis and risk assessment.^{19,20} In high-risk HPV-positive individuals, aberrant methylation of select genes (such as *PAX1*, *JAM3*, *SOX1*, *EPB41L3*, *DCC*, and *HS3ST2*²¹⁻²³) is closely linked to cervical carcinogenesis, offering a molecular basis for novel screening and triage strategies. Liang et al²⁴ reported that the combined detection of *PAX1* and *JAM3* methylation achieved 100% positivity for detecting cervical cancer within a high-risk HPV-positive cohort. Furthermore, in patients with non-type 16/18 high-risk HPV, a positive *PAX1m*/*JAM3m* test confers a CIN3+ risk of 39.1%, while a negative result reduces the risk to 0.9%.²¹ Building upon this evidence, this study focused on *PAX1m*/*JAM3m* as a triage biomarker in high-risk HPV-

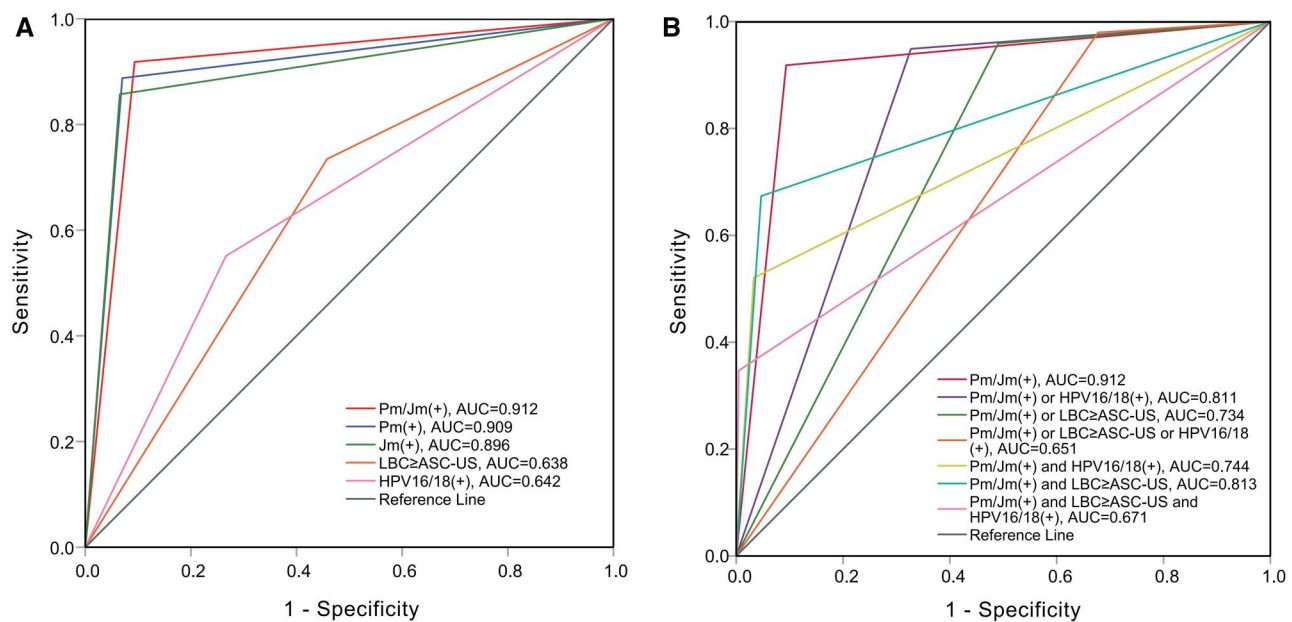


Figure 2 (A) Receiver operating characteristic curves of various methods for distinguishing CIN2+ lesions. (B) Receiver operating characteristic curves of combined methods for distinguishing CIN2+ lesions. AUC indicates area under the curve; CIN, cervical intraepithelial lesion; HPV-16/18(+), positivity for human papillomavirus 16 and/or 18; LBC ≥ ASC-US, liquid-based cytology test results atypical squamous cells of undetermined significance or worse; Pm(+), *PAX1m*(+), $\Delta Ct_{PAX1} \leq 6.6$; Jm(+), *JAM3m*(+), $\Delta Ct_{JAM3} \leq 10.0$; Pm/Jm(+), *PAX1m/JAM3m*(+), positivity for dual-gene *PAX1* or *JAM3* methylation.

Table 3 Effectiveness of *PAX1m/JAM3m* for colposcopy referral.

	Cases, No.	Colposcopy referral rate, % (n/No.)	CIN2+ detection rate, % (n/No.)	Referrals needed per CIN2+ case (No./n)	CIN2+ lesion miss rate, % (n/No.)
All high-risk HPV–positive					
HPV-16/18 positive	111	35.6 (111/312)	48.6 (54/111)	2.1 (111/54)	44.9 (44/98)
Liquid-based cytology ≥ AS-CUS	170	54.5 (170/312)	42.4 (72/170)	2.4 (170/72)	26.5 (26/98)
<i>PAX1m/JAM3m</i> positive	110	35.3 (110/312)	81.8 (90/110)	1.2 (110/90)	8.2 (8/98)
HPV 12 others positive					
Liquid-based cytology ≥ AS-CUS	121	60.2 (121/201)	30.6 (37/121)	3.3 (121/37)	15.9 (7/44)
<i>PAX1m/JAM3m</i> positive	52	25.9 (52/201)	75.0 (39/52)	1.3 (52/39)	11.4 (5/44)

Abbreviations: AS-CUS, atypical squamous cells of undetermined significance; CIN2+, cervical intraepithelial lesion grade 2 or higher; HPV, human papillomavirus; HPV other 12, any of the 12 additional high-risk human papillomavirus genotypes; *PAX1m/JAM3m*, dual-gene methylation of *PAX1* or *JAM3*.

positive individuals, aiming to provide an objective alternative to cytology, particularly suited for resource-constrained settings.

Our results demonstrated that *PAX1* and *JAM3* methylation positivity and levels increased with lesion severity. For the detection of CIN2+ lesions, *PAX1m/JAM3m* yielded a sensitivity of 91.8% (95% CI, 84.1%–96.2%), a specificity of 90.7% (95% CI, 85.7%–94.1%), and an AUC of 0.912 (95% CI, 0.874–0.951), outperforming liquid-based cytology ≥ AS-CUS and HPV-16/18 genotyping. This AUC value is notably higher than those reported for other methylation biomarkers, such as *EPB41L3* (AUC = 0.846)²⁵ and *SOX1* methylation (AUC = 0.611)²⁶ in similar patient populations. This superior discriminative ability of *PAX1m/JAM3m*, compared with other methylation markers, is further corroborated by previous research. For instance, Jing et al²⁷ reported that in HPV-16/18–positive women, *PAX1m/JAM3m* achieved a sensitivity of 89.0% (95% CI, 82.9%–95.1%), a specificity of 95.3% (95% CI, 92.6%–98.0%), and an AUC of 0.921 (95% CI,

0.877–0.966) for CIN2+ lesions, identifying all carcinoma cases. In women with AS-CUS, Chen et al²⁸ found that *PAX1m/JAM3m* complemented high-risk HPV testing, with a sensitivity of 83.8% and a specificity of 95.8% for CIN2+ lesions. Despite differences in study populations, these findings collectively support the utility of *PAX1m/JAM3m* in identifying high-grade lesions and reinforce its potential as a robust triage tool for abnormal cervical screening results. We further evaluated the performance of *PAX1m/JAM3m* in combination with cytology or HPV-16/18 genotyping. Consistent with a recent multicenter prospective study,²⁹ we found that although combining biomarkers slightly improved sensitivity for CIN2+ lesion detection, it came at the cost of markedly reduced specificity and PPV. Critically, these combinations yielded no clinically significant improvement in the AUC. These results indicate that *PAX1m/JAM3m* alone provides superior triage performance in settings where diagnostic resources are constrained.

Colposcopy remains the definitive diagnostic tool following abnormal cervical screening, but referring individuals who do

not have high-grade lesions increases the risk of overtreatment, psychological burden, and unnecessary use of resources,²⁷ which has prompted the exploration of methylation-based triage in optimizing referral strategies. A multicenter study reported that *FAM19A4/miR-124-2* methylation testing reduced colposcopy referrals by approximately 34% while maintaining detection of all cervical cancers and at least 70% of CIN3+ lesions.³⁰ In this study, using *PAX1m/JAM3m* positivity as a triage criterion reduced the overall colposcopy referral rate from 54.5% (liquid-based cytology $\geq$ AS-CUS) to 35.3% while increasing the CIN2+ lesion detection rate from 42.4% to 81.8%, representing a 19.2% reduction in unnecessary referrals and a 39.4% increase in detection efficiency. In the HPV-16/18-negative subgroup, *PAX1m/JAM3m* reduced the colposcopy referral rate from 60.2% to 25.9% and improved the CIN2+ lesion detection rate from 30.6% to 75.0%. The remarkable efficiency of our strategy is strongly supported by the work of Shang et al,³¹ who demonstrated that *PAX1m/JAM3m* triage reduced the number of colposcopy referrals needed to detect 1 CIN3+ case from 11.1 to just 2.5. Together, these findings underscore the transformative potential of *PAX1m/JAM3m* triage in enhancing the precision and efficiency of cervical screening programs.

Collectively, our findings provide a robust foundation for the clinical translation of *PAX1m/JAM3m* as a triage tool for high-risk HPV-positive women. The primary value of this strategy lies in its ability to substantially reduce unnecessary colposcopy referrals while maintaining high sensitivity for high-grade lesions, thereby alleviating patient anxiety, minimizing overtreatment, and optimizing health care resource allocation. From a health economics perspective, although the direct assay cost may be higher than for cytology, the strategy is highly likely to be cost-saving or cost-effective at the system level due to the avoidance of a large number of additional colposcopies and consequent biopsies. Furthermore, the objective and reproducible nature of this molecular test effectively eliminates the subjectivity and high expertise dependency inherent in cytologic interpretation. This key attribute makes it particularly suited for standardized implementation in diverse clinical settings, including those with limited cytologic expertise.

To translate these advantages into routine practice, validation in large-scale, prospective, and multicenter screening cohorts remains essential to establish long-term reproducibility, assess population-level effectiveness, and provide the robust evidence base required for incorporation into national and international clinical guidelines. This transition will require a concerted effort on 3 fronts: (1) the establishment of standardized protocols for sample collection, transport, and analysis to ensure consistent performance; (2) the development of automated platforms to streamline labor-intensive steps, such as bisulfite conversion and nucleic acid extraction, thereby improving efficiency and reducing turnaround time; and (3) the formulation of evidence-based clinical guidelines for interpreting results, with specific recommendations for managing cases where methylation findings are discordant with histopathology.

This study has several limitations: (1) The cohort was limited to high-risk HPV-positive individuals who had undergone histopathologic evaluation, introducing potential selection bias because some individuals either did not meet

colposcopy referral criteria or declined biopsy; (2) the absence of longitudinal follow-up data for *PAX1m/JAM3m*-positive individuals with histopathologic findings below CIN2 restricts assessment of clinical outcomes in this subgroup; and (3) the relatively small sample size underscores the need for validation in larger, prospective clinical studies.

In conclusion, this study demonstrates that *PAX1m/JAM3m* levels vary substantially across cervical lesion grades, with higher methylation levels correlating with increasing lesion severity. The diagnostic performance of *PAX1m/JAM3m* alone in detecting high-grade cervical lesions is comparable to or surpasses that of cytology or HPV genotyping, supporting its potential as an independent triage tool for high-risk HPV-positive populations.

Acknowledgments

The authors are grateful to all who took part in this study.

Conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research received funding from the Sichuan Province Medical Science Research Project Program (S23047) and the Health Commission of Sichuan Province Medical Science Technology Program (24WSXT093). The funders had no role in the design, data collection, data analysis, or reporting of this study.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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